Victor Angel Lopez Romero

Software Developer | AI Engineer

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Irapuato, Guanajuato, Mexico | Target location: Leon, Guanajuato

Professional summary

Software developer specializing in AI engineering, with over one year of professional experience building conversational agents, RAG systems, and LLM workflows for a business platform. Focused on understanding how users work and applying AI to save time, simplify tasks, and reduce costs.

Work experience

Vortice Coaching Oct 2025 - Present
AI Software Engineer

- Built a conversational AI agent with tool calling to retrieve information through application APIs.
- Implemented RAG features for question answering based on documents, with semantic retrieval and source references.
- Developed LLM workflows for information analysis, summarization, and structured outputs.
- Built backend services and APIs with input validation, access control, and data persistence.
- Implemented process management features and integrations across services. Contributed to interface and interaction improvements.
- Reduced token consumption by 42% against the existing baseline, measured in development tests.

Academic projects

TourlyAI: NLP for tourism reviews | github.com/TourlyAI Sep 2025 - Mar 2026
Academic project | *Software development and model fine-tuning*

- Built an NLP pipeline with 9 stages for tourism reviews, combining sentiment analysis, BERTopic topic modeling, and summaries and insights generated by LLMs.
- Fine-tuned and evaluated 2 BERT models with PyTorch and Hugging Face: multi-label classification across 12 categories (80% weighted F1) and subjectivity detection (77% F1).
- Integrated the pipeline into an Electron, React, and Python desktop app with interactive dashboards and PDF reports, using LangChain for local inference with Ollama and optional OpenAI integration.

Skills

Generative AI: RAG, AI agents, tool calling, LangChain, embeddings, vector databases, structured outputs, OpenAI, Ollama

ML and NLP: PyTorch, Hugging Face, scikit-learn, pandas, NumPy, BERT fine-tuning, model evaluation, BERTopic

Programming and APIs: Python, TypeScript, JavaScript, Node.js, Express, REST APIs

Data and messaging: SQL, PostgreSQL, Qdrant, Redis, BullMQ

Frontend and tooling: React, Electron, Docker, Git

Familiar with: Context engineering, prompt engineering, multi-agent patterns (agents-as-tools, swarm), agent evaluation, LangGraph, MCP, A2A

Education

Universidad de Guanajuato Aug 2023 - Nov 2026 (expected completion)
B.Eng. in Data Engineering & Artificial Intelligence (in progress)

Certifications

AWS Certified Developer - Associate Oct 2026
AWS Certified AI Practitioner Nov 2026

Languages

Spanish: Native | English: B1 (Intermediate)